DRIP PROTOCOL

Encrypted Sports Finance on EVM

Fhenix Privacy-by-Design Buildathon — Project Description

Inspired by the Stellar Drip sports finance architecture | Built on Fhenix + EVM

1. Project Overview

Drip Protocol is a fully on-chain sports finance protocol built on the Fhenix EVM (deployed on Arbitrum Sepolia). It takes direct architectural inspiration from Stellar Drip — a Soroban-native sports finance design — and ports it to the EVM, adding a full Fully Homomorphic Encryption (FHE) privacy layer that makes the financial stack viable for serious players and institutions.

The core thesis: a football simulation protocol generates real, verifiable on-chain data (match results, win/loss records) that powers a layered DeFi stack — betting pools, fan token bonding curves, and a perpetuals vAMM. Transparency on the game layer builds trust. Encryption on the financial layer prevents exploitation.

Why reference Stellar Drip?

Stellar Drip established the cleanest architecture for sports-native DeFi: Club NFTs as the primitive, tactical match execution as the oracle, fan tokens as community equity, and perps as the financial instrument. Drip Protocol preserves this architecture exactly — and makes it production-viable on EVM by solving the front-running and liquidation-hunting problems that make transparent financial rails unusable for real capital.

2. The Problem — Why Transparent Rails Break Sports Finance

On-chain sports betting and perpetuals have failed to attract serious capital not because the game mechanics are wrong, but because the financial mechanics are publicly exploitable:

- Parimutuel betting pools expose aggregate bets in real time. Large bettors are front-run before settlement.

- Perpetual positions — size, leverage, and liquidation price — are fully visible. Adversarial liquidators push the vAMM price toward the most exposed cluster.
- Fan token trades on bonding curves are mempool-visible. Whale buys are sandwich-attacked before the price adjusts.
- Open interest driving the funding rate is public. Sophisticated actors game the rate calculation by watching OI tilt.

The result: retail users lose to bots; institutional capital never enters; the protocol captures fees from a shrinking user base.

The Architectural Decision

Drip Protocol treats this as an architectural problem, not a UI problem. The solution is not rate limiting, not commit-reveal schemes with trusted coordinators, and not L2 sequencer tricks. It is FHE — computation on encrypted state, settled with on-chain guarantees, no trusted intermediary.

3. The Solution — Selective Encryption Architecture

Drip Protocol encrypts the financial layer and keeps the game layer transparent. This is a deliberate architectural choice, not a limitation.

3.1 What stays transparent

- Club NFT stats (Attack, Defense, Midfield, Stamina) — verifiable by all
- Match execution and PRNG — fair outcomes provable on-chain
- Win/loss records — public performance history
- Fan token bonding curve price function — anyone can verify the pricing formula
- Protocol fee accrual — transparent treasury

3.2 What gets encrypted via FHE

- Individual bet amounts and sides — sealed until match settlement
- Perpetual position size, leverage, and liquidation price — hidden from liquidation hunters
- Aggregate open interest used for funding rate computation — rate settles correctly on encrypted state
- Fan token trade size — whale buys and sells cannot be front-run
- Pre-match tactical formation selection — sealed until kick-off, enabling genuine game theory

Feature	Without FHE	With FHE (Drip Protocol)
Betting pool	Bet amounts public; odds manipulable	Sealed bets; parimutuel settled via FHE arithmetic
Perp positions	Liq prices visible; position hunting trivial	Encrypted positions; protocol proves undercollat. via FHE
Fan token trades	Sandwich attacks on large trades	Trade size private until settlement
Funding rate OI	OI visible; rate front-runnable	Aggregate OI computed on ciphertext
Tactical formations	Copyable before match starts	Sealed reveal at kick-off — new game mechanic

4. Protocol Features

4.1 Club NFT System

Club NFTs are the primitive that anchors the entire protocol. Each NFT represents a football club with four on-chain attributes — Attack, Defense, Midfield, Stamina — randomised at mint using Fhenix-compatible PRNG. Clubs accumulate win/loss records on-chain and can be upgraded by burning $DRIP tokens.

- Mint with on-chain randomised stats (10–100 per attribute)
- Upgrade attributes by spending $DRIP — cost formula: base_cost x (level+1)^2 / 10
- Upgrade contract verifies ownership via cross-contract call before charging
- Win/loss records stored on-chain and used by the fan token unlock condition
- 1% protocol fee on upgrades, withdrawable by admin

4.2 Tactical Match System

Matches are the oracle that generates all downstream financial data. The match engine is deterministic given club stats and a PRNG seed, making results verifiable and manipulation-resistant.

- Challenge mechanics: challenger owns club_1, acceptor owns club_2
- Outcome determined by club stats weighted against Fhenix PRNG — not manipulable by validators
- Match contract cross-calls Club NFT contract to read live stats
- Results trigger: fan token price movement, perp index settlement, betting pool resolution
- NEW: Pre-match formation selection submitted as FHE-encrypted inputs — revealed atomically at kick-off

4.3 Sealed-Bid Betting

The betting pool is re-architected from a transparent parimutuel to a sealed-bid pool using FHE. Bettors submit encrypted amounts and sides. The tally of bets-per-side is computed on encrypted state. Odds are only calculable at settlement time — making pre-settlement manipulation impossible.

- Encrypted bet size and side submitted via Fhenix SDK
- Pool tally computed over ciphertexts — no one can read running odds
- Draws refund all bettors automatically
- 2.5% protocol fee on winning pools, withdrawable by admin
- Settlement triggered exclusively by the match contract — access-controlled

4.4 Fan Tokens (Encrypted Bonding Curve)

Clubs that win 10 matches unlock their fan token. The bonding curve pricing formula is public, but individual trade sizes are encrypted — preventing front-running of large trades that would destabilise early-stage curves.

- Linear bonding curve: price rises with every buy, falls with every sell
- Trade size submitted encrypted — price impact settled on-chain after FHE computation
- 3% royalty on every buy/sell — flows to club owner
- 1% protocol fee on every buy/sell
- Slippage protection maintained even with encrypted trade sizes
- Instant liquidity — sell back to curve at any time

4.5 Perpetuals (Encrypted vAMM)

The perpetuals market is the most sophisticated component and the clearest FHE use case. The virtual AMM maintains a win-probability index (0–100%) for each club, settable by match outcomes. Positions are stored encrypted — size, leverage, and liquidation price are never visible to potential adversaries.

- Trade club win-probability index with up to 10x leverage
- Virtual AMM (constant-product) — no real liquidity or counterparty needed
- Position data (size, leverage, liq price) stored as FHE ciphertexts
- Funding rate computed on encrypted aggregate OI — rate settles correctly without revealing book state

- Liquidation: protocol proves position is undercollateralised via FHE comparison before executing — liquidator earns 1% without seeing the position
- Match settlement moves index: win = 75%, draw = 50%, loss = 25%
- 0.1% taker fee on open and close

5. FHE Architecture — Technical Design

5.1 Encryption Stack

Drip Protocol uses the Fhenix CoFHE stack for all encrypted computation. The client-side SDK handles encryption before data hits the chain. Smart contracts operate on FHE ciphertext types natively.

Layer	Technology	Purpose
Smart contracts	Solidity + FHE library (euint32, ebool)	Encrypted state storage and computation
Client SDK	@cofhe/sdk	Client-side encrypt/decrypt, permit generation
React hooks	@cofhe/react (useEncrypt, useDecrypt)	Frontend integration
Local dev	Hardhat plugin (@cofhe/hardhat)	FHE simulation without testnet
Testnet	Arbitrum Sepolia (Fhenix CoFHE)	Live encrypted compute layer
Token standard	ERC-721 (Club NFTs), ERC-20 (fan tokens, $DRIP)	Standard EVM composability
vAMM	Custom constant-product, encrypted OI	Perp settlement with private state

5.2 Encrypted Data Types

- euint32 — bet amounts, position sizes, trade sizes, OI
- ebool — bet side (long/short), match outcome gates
- euint8 — formation selection (tactical input, 0–255 strategy index)
- All types handled natively by Fhenix CoFHE Solidity library

5.3 Access Control Model

Decryption permissions follow the Fhenix permit system. Users can decrypt their own position data. The protocol can decrypt aggregate values (total OI, tally per side) for settlement computation. No trusted third party holds decryption keys.

- User permit: owner can view own position, own bet, own token balance in plaintext
- Protocol settlement: aggregate decryption gated behind match contract access control
- Liquidation trigger: protocol calls FHE comparison (position_equity < threshold) — boolean result only, no size revealed

6. $DRIP Token

- Utility token — spent to upgrade club attributes (burns tokens, deflationary pressure)
- Earned through match wins — protocol mints $DRIP to winning club owner
- Required to challenge matches above a certain tier — skin-in-the-game mechanic
- Protocol treasury funded by all fee streams (betting, upgrades, fan tokens, perps)

7. Full Technology Stack

Layer	Technology	Purpose
EVM runtime	Fhenix CoFHE on Arbitrum Sepolia	Encrypted compute layer
Smart contracts	Solidity 0.8.x + @fhenixprotocol/contracts	All protocol logic
Dev tooling	Hardhat + @cofhe/hardhat plugin	Local FHE simulation
Frontend	Next.js + wagmi + @cofhe/react	User-facing dApp
Backend indexer	The Graph (subgraph)	Match history, position indexing
Wallet	MetaMask / WalletConnect (EVM-standard)	Standard EVM UX
Reference arch	Stellar Drip (Soroban)	Game layer design inspiration

8. Buildathon Category Fit

Drip Protocol is submitted under two intersecting categories from the Fhenix buildathon brief:

- Confidential DeFi — sealed-bid betting pools, encrypted perp positions, MEV-protected bonding curve trades
- Encrypted Gaming — hidden tactical formations, private player actions, provably fair match outcomes

The intersection is the core novelty: the game generates the oracle data; the DeFi stack consumes it. Both layers must exist for either to work. FHE is what makes the financial layer viable rather than exploitable.

9. Development Milestones Summary

Wave	Phase	Deliverable	Grant Target
Wave 1	Foundation	Club NFT + Match Engine contracts deployed on testnet	$3,000
Wave 2	Financial Layer	Betting pool (sealed-bid FHE) + Fan tokens live	$5,000
Wave 3	Core Protocol	Encrypted vAMM perps + $DRIP token + full integration	$12,000
Wave 4	Frontend + UX	Complete dApp, subgraph indexer, encrypted UX flows	$14,000
Wave 5	Audit + Launch	Security review, mainnet prep, ecosystem showcase	$16,000

10. Team

Drip Protocol is submitted as a solo build. The founder brings a track record across multiple EVM and non-EVM hackathons, including a first-place finish at the Seedify Fund BNB Chain Hackathon (PredictBNB — prediction market protocol). Prior work spans DeFi primitives, oracle systems, NFT infrastructure, and on-chain gaming across Solana, Arbitrum, BNB Chain, Sui, and Stacks.

Relevant prior art: IVirtualz (virtual Premier League betting on Arbitrum, Chainlink VRF), PredictBNB (prediction markets, BNB Chain), ProofPlay (gaming oracle on Sui), NeutralCurveAMM (DEX infrastructure), PayFiEscrow (stablecoin payment rails).